

3.1 Discrete Random Variables

Given is a random experiment E (e.g. ruling out a target in a maximum of five attempts) with a corresponding sample space S

$$S = \{ h, mh, mmh, mmmh, mmmmh \}$$

where m = miss and h = hit.

Let X = the number of the attempt at which the target is ruled out.

Since the range of $X = \{1, 2, 3, 4, 5\}$, X has a finite range. So X is a discrete random variable.

3.2/3.3 Probability Mass Functions & Cumulative Distribution Functions

Definition

probability mass function:	$f(x)$	=	$P(X = x)$
cumulative distribution function:	$F(x)$	=	$P(X \leq x)$

Example 1

Ruling out a target in a maximum of five attempts. Let X = the number of the attempt at which the target is ruled out. Suppose $P(\text{hit at attempt } i \mid \text{miss at attempts } 1 \text{ to } i-1) = i/5$.

Determine the probability distribution of X (i.e. a table with all possible values of $f(x)$) and determine the cumulative distribution function. Draw the graphs of $f(x)$ and $F(x)$.

Solution

$$\begin{aligned}
 P(X=1) &= P(h_1) \\
 &= 1/5 &= 0.2 \\
 \\
 P(X=2) &= P(m_1h_2) \\
 &= P(m_1) \cdot P(h_2 \mid m_1) \\
 &= (1 - 1/5) \cdot 2/5 &= 0.32 \\
 \\
 P(X=3) &= P(m_1m_2h_3) \\
 &= P(m_1) \cdot P(m_2 \mid m_1) \cdot P(h_3 \mid m_1m_2) \\
 &= (1 - 1/5) \cdot (1 - 2/5) \cdot 3/5 &= 0.288
 \end{aligned}$$

So

Table 3.1 The probability distribution of Example 1

x	1	2	3	4	5
$f(x)$	0.2	0.32	0.288	0.1536	0.0384

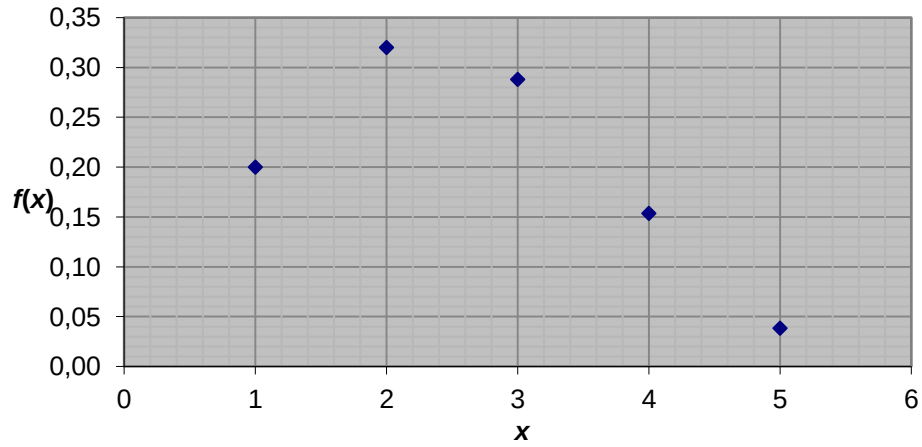


Figure 3.1 Probability mass function for Example 1

$$\begin{aligned}
 F(2) &= P(X \leq 2) &= P(X = 1 \vee X = 2) \\
 &= P(X = 1) + P(X = 2) &= 0.2 + 0.32 &= 0.52 \\
 F(3\frac{1}{2}) &= P(X \leq 3\frac{1}{2}) &= P(X = 1 \vee X = 2 \vee X = 3) \\
 &= P(X = 1) + P(X = 2) + P(X = 3) &= 0.2 + 0.32 + 0.288 &= 0.808 \\
 F(5) &= P(X \leq 5) &= 1
 \end{aligned}$$

So

$$F(x) = \begin{cases} 0 & (x < 1) \\ 0.2000 & (1 \leq x < 2) \\ 0.5200 & (2 \leq x < 3) \\ 0.8080 & (3 \leq x < 4) \\ 0.9616 & (4 \leq x < 5) \\ 1 & (x \geq 5) \end{cases}$$

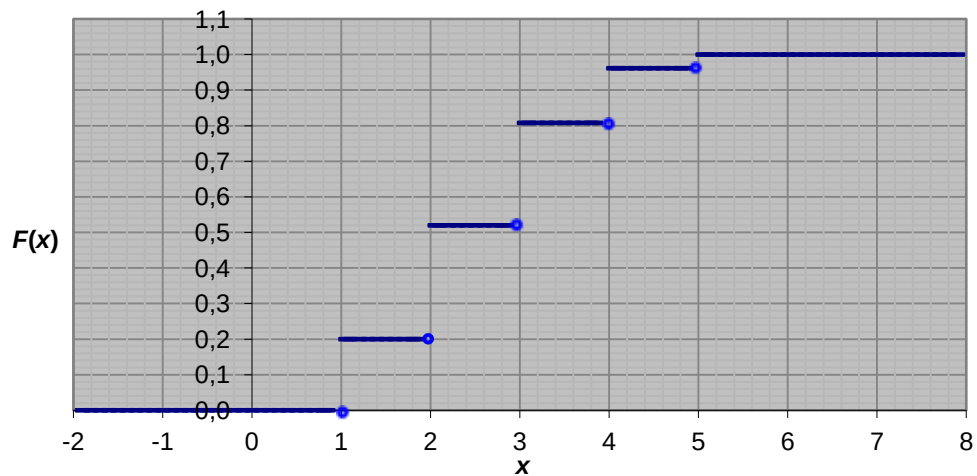


Figure 3.2 Cumulative distribution function of Example 1

Remarks

- $f(x) \geq 0$ ($\forall x$) and $\sum f(x) = 1$
- $F(x)$ is nondecreasing, $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$

3.4 Mean and Variance

Analogously to the concepts **population mean** and **population variance** from the descriptive statistics, the **expected value** and the **variance** of a discrete random variable X are defined.

Definition

$$\begin{aligned}\mu &= E(X) = \sum xf(x) \\ \sigma^2 &= V(X) = E((X - E(X))^2) = E(X^2) - E(X)^2\end{aligned}$$

Example 2

See the data in Example 1. Determine $E(X)$ and $V(X)$.

Solution

$$E(X) = 1 \cdot 0.2 + 2 \cdot 0.32 + 3 \cdot 0.288 + 4 \cdot 0.1536 + 5 \cdot 0.0384 = 2.5104$$

$$\begin{aligned}V(X) &= (1 - 2.5104)^2 \cdot 0.2 + (2 - 2.5104)^2 \cdot 0.32 + \dots + (5 - 2.5104)^2 \cdot 0.0384 \\ &= 1.18749184 \approx 1.1875\end{aligned}$$

Alternative method

$$E(X^2) = 1^2 \cdot 0.2 + 2^2 \cdot 0.32 + 3^2 \cdot 0.288 + 4^2 \cdot 0.1536 + 5^2 \cdot 0.0384 = 7.4896$$

$$V(X) = E(X^2) - E(X)^2 = 7.4896 - 2.5104^2 = 1.1875$$

So the **standard deviation** σ is equal to $\sqrt{1.1875} = 1.0897$.

Properties of $E(X)$ and $V(X)$

$$\begin{aligned}E(a \cdot X + b) &= a \cdot E(X) + b \\ V(a \cdot X + b) &= a^2 \cdot V(X)\end{aligned}$$

Example 3

See the data in Example 1. Determine the expected value and the standard deviation of the cost function C , if there are € 2,000 fixed costs and € 1,000 variable costs per attempt.

Solution

Since $C = 1000 \cdot X + 2000$, $E(C) = 1000 \cdot E(X) + 2000 = € 4,510.40$.

Since $V(C) = 1000^2 \cdot V(X) = 1\,187\,491.84$, $\sigma = € 1,089.72$.

3.5 Discrete Uniform Distribution

A random variable X has a **discrete uniform distribution** if each of the n values in its range $x_1, x_2, \dots, x_n$ has equal probability, i.e.

$$P(X = x_i) = 1/n \quad (i = 1, 2, 3, \dots, n).$$

Example 4

Tossing with a fair die. Let X = the number of dots. Then $P(X = i) = 1/6$ ($i = 1, 2, \dots, 6$).

Properties of $E(X)$ and $V(X)$

Suppose the range of the discrete uniform random variable X consists of n consecutive integers: $a, a+1, a+2, \dots, b = a+n-1$. Then

$$\begin{aligned} E(X) &= \frac{a+b}{2} \\ V(X) &= \frac{(b-a+1)^2-1}{12} \end{aligned}$$

Proof

$$\begin{aligned} E(X) &= \sum_{x=a}^b x \cdot P(X = x) \\ &= \sum_{x=a}^b x \cdot \frac{1}{n} \\ &= \frac{1}{n} \cdot (a + (a+1) + (a+2) + \dots + (a+n-2) + (a+n-1)) \\ &\quad \begin{array}{c} \text{Diagram showing the sum of an arithmetic series with } n \text{ terms. The first term is } a \text{ and the last term is } b. \text{ The sum is represented by } \frac{n}{2} \cdot (a+b). \end{array} \\ &= \frac{1}{n} \cdot \left(\frac{n}{2} \cdot (a+b) \right) \\ &= \frac{a+b}{2} \end{aligned}$$

Examples

5. Tossing with a fair die. Let X = the number of dots. So X is a discrete uniform random variable with $a = 1$ and $b = 6$. Determine $E(X)$ and $V(X)$.

Thus

$$\begin{aligned} E(X) &= (1 + 6)/2 = 3.5; \\ V(X) &= ((6 - 1 + 1)^2 - 1)/12 = 35/12. \end{aligned}$$

6. Thickness measurements X of a coating process are uniformly distributed with values 2.1, 2.2, 2.3, 2.4 and 2.5. Determine $E(X)$ and $\sigma(X)$.

Let $X = 0.1 \cdot Y$. Then Y **uniformly distributed** with $a = 21$ and $b = 25$.

Since $E(Y) = (21 + 25)/2 = 23$ and $V(Y) = ((25 - 21 + 1)^2 - 1)/12 = 2$, it follows

$$\begin{aligned} E(X) &= 0.1 \cdot 23 = 2.3; \\ V(X) &= 0.1^2 \cdot 2 = 0.02 \Rightarrow \sigma(X) = 0.1414. \end{aligned}$$